

AMBULONG, Philippines

WMO: 984320

Lat: 14.0877N

Lon: 121.0625E

Elev: 11

StdP: 101.19

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.8	22.8	19.0	13.8	23.5	19.7	14.5	24.1	7.0	25.8	6.3	26.8	1.4	40	0.332

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.5	34.9	28.2	34.2	28.0	33.6	27.9	29.4	33.2	29.0	32.8	28.6	32.3	2.2	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
28.3	24.6	32.4	27.9	24.0	31.9	27.6	23.5	31.4	96.6	33.5	94.5	32.9	92.7	32.5	35.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.0	5.3	4.7	19.5	36.8	2.5	1.4	17.7	37.9	16.2	38.7	14.8	39.5	13.0	40.5
			18.8	30.8	2.0	1.5	17.4	31.8	16.2	32.7	15.1	33.5	13.6	34.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	26.6	27.0	28.1	29.5	29.7	29.1	28.3	28.3	28.2	28.1	28.0	27.2
	DBStd	1.44	1.25	1.26	1.23	1.09	1.06	1.11	1.10	1.07	1.14	1.02	1.04	1.09
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6637	514	475	562	585	610	573	568	567	547	561	541	534
	CDD18.3	3595	256	242	303	335	352	323	309	309	297	302	291	276
	CDH23.3	39129	2131	2163	3257	4179	4461	3921	3419	3477	3227	3246	3069	2581
	CDH26.7	14201	488	645	1232	1966	2085	1649	1171	1189	1083	1099	974	619
(14) Wind	WSAvg	1.8	2.5	2.2	2.0	1.6	1.4	1.3	1.6	1.9	1.5	1.5	1.9	2.5
(15) Precipitation	PrecAvg	1900	29	10	15	36	134	274	314	296	282	199	192	114
	PrecMax	2513	160	53	45	105	521	540	1096	636	817	429	348	317
	PrecMin	1178	2	0	1	1	37	109	80	90	42	47	19	7
	PrecStd	443	45	15	14	31	130	138	253	142	173	138	103	94
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.9	33.0	34.6	35.7	35.8	35.1	33.8	33.7	33.7	33.3	33.1	32.1
		MCWB	26.8	26.7	27.7	28.1	28.6	28.8	28.7	28.0	28.5	28.0	27.9	27.0
	2%	DB	30.9	31.8	33.4	34.9	34.8	34.0	32.8	32.6	32.6	32.5	32.1	31.2
		MCWB	26.2	26.3	27.2	27.7	28.4	28.6	28.2	28.0	28.1	27.5	27.3	26.6
	5%	DB	30.1	31.0	32.6	34.2	34.1	33.2	32.0	31.7	31.8	31.8	31.5	30.5
		MCWB	25.7	25.7	26.8	27.6	28.2	28.4	27.7	27.6	27.7	27.1	26.9	26.2
	10%	DB	29.3	30.2	31.8	33.5	33.2	32.4	31.2	30.9	31.0	31.1	30.8	29.7
		MCWB	25.2	25.4	26.4	27.3	28.0	28.1	27.4	27.3	27.4	26.9	26.6	25.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.5	28.5	29.2	29.6	30.0	29.7	29.5	29.2	29.5	29.0	28.7	27.8
		MCDB	30.9	31.3	32.8	34.3	34.1	33.9	32.8	32.4	32.5	32.3	32.2	31.0
	2%	WB	26.6	27.3	28.2	28.7	29.2	29.1	28.7	28.5	28.6	28.2	27.9	27.1
		MCDB	30.2	30.6	32.3	33.6	33.4	33.0	31.9	31.6	31.7	31.5	31.4	30.4
	5%	WB	26.1	26.4	27.4	28.2	28.7	28.7	28.2	28.0	28.1	27.7	27.4	26.6
		MCDB	29.5	30.0	31.8	33.1	32.8	32.4	31.3	30.9	31.2	30.9	30.7	29.7
	10%	WB	25.5	25.7	26.7	27.7	28.2	28.2	27.7	27.5	27.6	27.2	26.9	26.1
		MCDB	28.6	29.3	31.1	32.5	32.2	31.8	30.5	30.2	30.5	30.2	30.1	29.0
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.9	7.6	8.4	7.5	6.6	5.7	5.3	5.7	6.2	5.9	5.0
		MCDBR	6.7	8.2	8.9	9.5	8.6	7.8	7.2	6.9	7.1	7.4	7.0	6.4
		MCWBR	3.4	3.9	4.1	4.1	3.8	3.7	3.6	3.4	3.6	3.5	3.2	3.1
	5% WB	MCDBR	6.4	7.2	8.5	8.9	7.8	7.4	6.7	6.2	6.7	6.7	6.6	5.9
		MCWBR	3.6	4.1	4.5	4.4	4.1	4.0	3.8	3.4	3.7	3.7	3.4	3.1
(40) Clear-Sky Solar Irradiance	taub		0.427	0.423	0.450	0.467	0.467	0.463	0.459	0.466	0.462	0.477	0.443	0.447
	taud		2.396	2.398	2.325	2.281	2.310	2.339	2.349	2.320	2.325	2.263	2.371	2.342
	Ebn at Noon		869	889	870	848	835	830	835	838	847	827	847	838
	Edh at Noon		117	121	133	138	132	127	126	131	131	136	119	121
(44) All-Sky Solar Radiation	RadAvg		4.26	5.06	5.71	6.38	5.76	5.17	4.60	4.48	4.69	4.52	4.26	3.80
	RadStd		0.29	0.46	0.41	0.35	0.52	0.48	0.57	0.42	0.48	0.35	0.38	0.34

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	+0.31	+0.49	N/A	+0.45	+0.37	+0.47	N/A	N/A	+158	+158

Nomenclature: See separate page